

Ivar Cashin Eriksson

Curriculum Vitae

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Introduction

Machine learning researcher and engineer with seven years of experience developing machine learning methods and software in research and industry. MSc in Engineering Physics, specialising in Statistical Learning and Data Analytics, as well as a supplementary BSc. in Economics. Research experience in probabilistic modelling and numerical optimisation, as well as earlier experience in automated financial trading. Looking to return to research with a focus on machine learning.

Education

- 2015–2020 **KTH, Royal Institute of Technology – MSc. in Engineering Physics**, GPA: 5.0 / 5.0
Specialisation: Applied Mathematics, Statistical Learning and Data Analytics.
Relevant coursework: Reinforcement Learning, Optimisation, Operations Research.
Thesis: [Probabilistic Dose Prediction for Radiation Therapy Planning](#)
- 2019 **ETH Zürich, Swiss Federal Institute of Technology – Applied Mathematics**, Exchange
Relevant coursework: Game Theory and Control, Computational Statistics, Applied Time Series.
- 2017–2022 **SU, Stockholm University – BSc. in Economics**, Additional Bachelor's Degree
- 2017 **ZJU, Zhejiang University**, Research exchange, Laser trimming of microring resonators.

Selected Work Experience

Researcher within Data Science @ RaySearch Laboratories

- Redesigned optimisation algorithms to reflect clinical decision hierarchies, validated with clinicians.
- Developed autoencoder-based patient embeddings and learned similarity from segmented 3D CT scans.
- Predicted dose-volume histograms and spatial dose distributions using sparse Gaussian processes.

Keywords: deep learning, representation learning, probabilistic modelling, Gaussian processes, numerical optimisation, clinical collaboration

[Read more...](#)

Data Scientist / ML Engineer @ Signum Life Science

- Develop automated ML workflows for pharmaceutical forecasting, from data validation and training to experiment tracking, batch inference, and reproducible evaluation.
- Design maintainable ML architecture supporting thousands of independently trained models.

Keywords: Python, machine learning, data pipelines, model evaluation, Databricks, MLflow, Docker, Linux

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Principal Data Scientist @ Valcon

- Built and deployed a real-time neural-network pipeline combining natural language and categorical inputs, using transfer learning to predict high-dimensional categorical outputs.
- Led ML projects across energy, IoT, and manufacturing, from problem formulation to deployment.
- Mentored junior data scientists and delivered training in explainable AI, Git, and software architecture.

Keywords: real-time ML, neural networks, production deployment, AI/ML architecture, technical leadership

[Read more...](#)

Skills and Other

Technology Python, PyTorch, TensorFlow, SQL, Git, Linux, Docker, MLflow, Databricks
Methods Optimisation, probabilistic modelling, deep learning, forecasting, model evaluation
Language Native English and Swedish, intermediate Danish

Detailed Work Experience

2024– Present **Data Scientist / ML Engineer, Signum Life Science, Copenhagen**

At Signum, I work across ML, software architecture, and project management, translating loosely defined problems in pharmaceutical analytics into maintainable production systems.

I lead architecture and development of a forecasting platform for pharmaceutical prices, demand, and patient populations. The platform combines data validation, model training, experiment tracking, and batch inference in Databricks, supporting thousands of independently trained models. My work includes reproducible evaluation, maintainable software design, and coordination with technical and domain stakeholders.

I also develop embedding-based retrieval and matching workflows for domain-specific text. This involves evaluating semantic similarity, analysing retrieval failures, and improving matching reliability.

2022–2024 **Principal Data Scientist, Valcon, Copenhagen**

At Valcon, I led ML projects across energy, IoT, pharmaceutical, and manufacturing domains, taking projects from problem formulation and model development through production deployment. I also mentored junior colleagues and delivered training in explainable AI, Git, and software architecture.

For one project, I designed and deployed an end-to-end pipeline for a real-time neural network. Using transfer learning, the model combined natural-language and high-dimensional categorical inputs to predict high-dimensional categorical outputs. I was responsible for the training pipeline, modular model architecture, evaluation, and production integration.

I received the Valcon People's Choice Award for exemplary work, leadership, team spirit, and scientific curiosity.

2021–2022 **Data Scientist, Valtech, Copenhagen**

At Valtech, I developed ML solutions for e-commerce, including time-series forecasting and customer-behaviour modelling. I built forecasting pipelines using Prophet and applied clustering to behavioural data to support personalisation and customer segmentation.

I worked with technical and commercial stakeholders to translate modelling results into operational recommendations and presented findings to senior decision-makers.

2019–2021 **Researcher within Data Science, RaySearch Laboratories, Stockholm**

At RaySearch Laboratories, I researched ML and optimisation methods for automated radiotherapy planning. The work combined deep learning, probabilistic modelling, and multi-objective optimisation in a safety-critical clinical setting.

I developed a pipeline that learned latent anatomical representations from segmented 3D CT scans using autoencoders. These representations were used to identify similar patients through a learned similarity metric and to predict dose-volume histograms and spatial dose distributions using sparse Gaussian processes.

I also redesigned lexicographic optimisation algorithms to better represent clinical decision hierarchies. I collaborated with medical physicists and clinicians to define requirements, evaluate model behaviour, and ensure that the resulting methods aligned with oncological standards and practical workflows.

2015–2018 **Trading Supervisor, Ampfield Aktiebolag AB, Stockholm**

Alongside my studies, I supervised automated trading systems. I also contributed C++ implementations of optimisation algorithms for trading strategies.

Personal Projects

Project iris End-to-end computer-vision retrieval system for making product images on e-commerce sites interactive. Built object detection, embedding generation, vector indexing, similarity search, backend APIs, persistence, dynamic scraping, and a Chromium extension using YOLOv8, OpenCLIP, Qdrant, FastAPI, and MongoDB. The repository can be found [here](#).